Week 1 Stepping Stones

DATA 422: Advanced Data Science Methods and Ethics

Section A: AI, ML, and Models

1. Who coined the term *machine learning* and in what year?
   Reading: §1.1
2. Briefly distinguish *AI* from *machine learning* as the book frames them.
   Reading: §1.1
3. Write the general model form $y = f(x)$. What do x, y, and f represent in supervised learning?
   Reading: §1.2

Section B: Four-Step ML Process

4. List the four steps of the supervised ML process.
   Reading: §1.3
5. In the house-price example, what are the **parameters** of the model, and what do they control?
   Reading: §1.2
6. Why do we need a **loss function**? Give one example appropriate to regression.
   Reading: §1.2–§1.3

Section C: Math Notation — Vectors & Matrices

7. Rewrite $y = wx + b$ for two features using vector notation.
   Reading: §1.4
8. Compute the dot product $w = [2, 3]$, $x = [4, 5]$.
   Reading: §1.4
9. Define **cosine similarity** between two vectors.
   Reading: §1.4 (Eq. 1.5)
10. **Cosine similarity practice (compute & draw):** For each pair,
    (i) compute cosine similarity, (ii) sketch the vectors
    (ii) say whether they're close in direction or far apart, and (iv) explain how the value reflects that.
    a) $[1, 0]$ and $[0, 1]$
    b) $[1, 0]$ and $[1, 1]$
    c) $[2, 2]$ and $[-2, -2]$
    d) $[3, 4]$ and $[6, 8]$

    Reading: §1.4
11. **Terminology check:** In this chapter, are "**parameters**" and "**weights**" used interchangeably? If so, what exactly are the parameters in $y = wx + b$?
    Reading: §1.2
12. **Notation check (labels vs. predictions):** In logistic regression, clearly define z, $\hat{y}$, and y.
    Which is the **logit**, which is the **model output**, and which is the **true label**?
    Reading: §1.7–§1.8

Section D: Logistic Regression & Gradient Descent

13. Write the logistic regression equation and the sigmoid definition.
    Reading: §1.7 (Eq. 1.8)
14. Why is sigmoid an appropriate activation for binary classification?
    Reading: §1.5, §1.7
15. Write the **binary cross-entropy** loss for one example.
    Reading: §1.7 (Eq. 1.9)
16. Why do we use **iterative** optimization (gradient descent) instead of a closed-form solution for logistic regression?
    Reading: §1.7
17. What does the **learning rate** control? What happens if it's too small or too large?
    Reading: §1.7

Section E: Tensors & Autograd

18. What is a **tensor** in PyTorch? *Reading: §1.8*
19. Why is **autograd** necessary for training neural networks at scale?
    Reading: §1.8
20. What does `loss.backward()` compute, and what does `optimizer.step()` do next?
    Reading: §1.8
    Reference: PyTorch Autograd docs

Section F: Mapping Equations PyTorch

21. Match each mathematical component to the PyTorch piece:
    - $\mathbf{w} \cdot \mathbf{x} + b$
    - $\sigma(z)$
    - binary cross-entropy *Reading: §1.8*
22. In:

    ```
    model = nn.Sequential(
        nn.Linear(2, 1),
        nn.Sigmoid()
    )
    ```

    what do the numbers `2` and `1` represent?
    Reading: §1.8
    Reference: PyTorch nn.Linear docs